Topic 1

The Group Axioms

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Definition 1: Let G be a set. A binary operation $*$ is a function $*\colon G \times G \to G$.

Notation: $x * y \coloneqq *(x, y)$.

Definition 2: Let G be a set and let $*$ be a binary operator on G. $(G, *)$ is said to be a group if the following three criteria are all met.

1. Associativity.
   - $\forall a, b, c \in G, (a * b) * c = a * (b * c)$.
2. Existence of an identity for $*$ in G.
   - $\exists e \in G$ such that $\forall a \in G, a * e = e * a = a$.
3. Existence of inverses for every element in G.
   - $\forall a \in G, \exists b \in G$ such that $a * b = b * a = a$.

Definition 3: Let $(G, *)$ be a group. $(G, *)$ is abelian if, in addition to the group axioms, it satisfies commutativity:

- $\forall a, b \in G, a * b = b * a$.

Exercise 1: Consider the operation of subtraction on $\mathbb{Z}$. Show by example that this operation is neither associative nor commutative. Additionally, show that there is no identity element for subtraction.

Solution:

(I) Show subtraction isn't associative on $\mathbb{Z}$.

Since:

$$\begin{aligned}(1-2)-3 = -1-3 = -4 \\ &\neq 2 = 1-(-1) = 1-(2-3)\end{aligned}$$

and $1, 2, 3 \in \mathbb{Z}$, subtraction isn't associative on $\mathbb{Z}$.

(II) Show subtraction isn't associative on $\mathbb{Z}$.

Since:

$$\begin{aligned} 0 - 1 &= -1 \\ &\neq 1 = 1 - 0 \end{aligned}$$

and $0, 1 \in \mathbb{Z}$, subtraction isn't commutative on $\mathbb{Z}$.

(III) Show subtraction doesn't have an identity element in $\mathbb{Z}$.

Let's assume, for the purposes of deriving a contradiction, that subtraction does have an identity element in $\mathbb{Z}$. Then there must exist some $e \in \mathbb{Z}$ such that:

$$x - e = e - x = x$$

for all $x \in \mathbb{Z}$. However, notice that:

$$\begin{aligned} 1 - e &= e - 1 \\ \implies 2e &= 2 \\ \implies e &= 1 \end{aligned}$$

However:

$$\begin{aligned} 0 - e &= e - 0 \\ \implies 2e &= 0 \\ \implies e &= 0 \end{aligned}$$

Hence, e must be equal to both 0 and 1 to be an identity element for subtraction over $\mathbb{Z}$, which clearly isn't possible. Therefore, e cannot exist and thus no identity element for subtraction over $\mathbb{Z}$ exists.

Exercise 2: Let $\mathbb{R}_{>0}$ denote the set of positive real numbers, and let $\wedge$ denote the exponentiation operation on $\mathbb{R}$; namely, $a \wedge b = a^b$. Is $\wedge$ associative? Commutative? Does it have an identity element?

Solution:

(I) Prove or disprove that $\wedge$ is associative over $\mathbb{R}_{>0}$.

$\wedge$ is <u>not</u> associative over $\mathbb{R}_{>0}$, since:

$$\begin{aligned} (2 \wedge 3) \wedge 2 = 2^3 \wedge 2 = 2^6 &= 64 \\ &\neq 512 = 2^9 = 2 \wedge 3^2 = 2 \wedge (3 \wedge 2) \end{aligned}$$

and $2, 3 \in \mathbb{R}_{>0}$.

(II) Prove or disprove that $\wedge$ is commutative over $\mathbb{R}_{>0}$.

$\wedge$ is <u>not</u> commutative over $\mathbb{R}_{>0}$, since:

$$\begin{aligned} 1 \wedge 2 = 1^2 = 1 \\ &\neq 2 = 2^1 = 2 \wedge 1 \end{aligned}$$

and $1, 2 \in \mathbb{R}_{>0}$.

(III) Prove or disprove the existence of an identity element for $\wedge$ over $\mathbb{R}_{>0}$.

$\wedge$ does not have an identity element in $\mathbb{R}_{>0}$. To prove this, let's assume for the purposes of deriving a contradiction that such an element $e \in \mathbb{R}_{>0}$ did exist. Then for arbitrary $a \in \mathbb{R}_{>0}$, we would have that:

$$a \wedge e = a$$

Implying $a^e = a$ and thus meaning e must be 1. However, e would also have to satisfy:

$$e \wedge a = a$$

to be a valid identity element. However, the equation above implies $e^a = a$ and since e was determined to be 1 earlier, we can would have to conclude that:

$$1^a = a$$

Which clearly isn't true for all $a \neq 1$. Thus, our assumption that e was an identity element for $\wedge$ over $\mathbb{R}_{>0}$ couldn't have been correct, and so we must conclude no identity element for $\mathbb{R}_{>0}$ under $\wedge$ exists at all.

Notation: $M_n(\mathbb{R})$ denotes the set of all $n \times n$ matrices with only real entries.

Exercise 3: Prove that matrix multiplication is associative, and give an example showing that matrix multiplication is not commutative.

Solution:

(I) Prove that matrix multiplication is associative.

Fix some $n \in \mathbb{N}$ and let $A, B, C \in M(\mathbb{R})$. Since:

$$
\begin{aligned}
((AB)\,C) &= \sum_{k=1}^{n} (AB)_{ik}\, C_{kj} \\
&= \sum_{k=1}^{n} \sum_{l=1}^{n} A_{il} B_{lk} C_{kj} \\
&= \sum_{l=1}^{n} A_{il}\,(BC)_{lj} \\
&= (A\,(BC))_{ij}
\end{aligned}
$$

Thus, since $((AB)\,C)_{ij} = (A\,(BC))_{ij}$, $(AB)\,C = A\,(BC)$ and therefore matrix multiplication is associative $M_n(\mathbb{R})$.

(II) Show matrix multiplication isn't commutative.

Since, $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in M_2(\mathbb{R})$, but:

$$
\begin{aligned}
\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} &= \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \\
&\neq \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}
\end{aligned}
$$

Matrix multiplication isn't generally commutative on $M_n(\mathbb{R})$.

Theorem 1: Let $A \in M_n(\mathbb{R})$. A is invertible if and only if $det(A) \neq 0$.

Exercise 4:

$$
\begin{pmatrix} 2 & 1 & 1 \\ 5 & 3 & 1 \\ 3 & 2 & 1 \end{pmatrix}, \begin{pmatrix} 2 & 1 & -1 \\ 5 & 3 & 1 \\ 3 & 2 & 2 \end{pmatrix}
$$

Solution:

(I) Check if the first matrix is invertible.

We know the first matrix is invertible if its determinant is 0. Since:

$$
\begin{aligned}
\begin{vmatrix} 2 & 1 & 1 \\ 5 & 3 & 1 \\ 3 & 2 & 1 \end{vmatrix} &= 2\begin{vmatrix} 3 & 1 \\ 2 & 1 \end{vmatrix} - \begin{vmatrix} 5 & 1 \\ 3 & 1 \end{vmatrix} + (-1)\begin{vmatrix} 5 & 3 \\ 3 & 2 \end{vmatrix} \\
&= 2(3-2) - (5-3) - (5(2) - 2(3)) \\
&= 2(1) - 2 - 1 \\
&= -1
\end{aligned}
$$

The matrix is therefore invertible.

(II) Check if the second matrix is invertible.

Since:

$$
\begin{aligned}
\begin{vmatrix} 2 & 1 & -1 \\ 5 & 3 & 1 \\ 3 & 2 & 2 \end{vmatrix} &= 2\begin{vmatrix} 3 & 1 \\ 2 & 2 \end{vmatrix} - \begin{vmatrix} 5 & 1 \\ 3 & 2 \end{vmatrix} - \begin{vmatrix} 5 & 3 \\ 3 & 2 \end{vmatrix} \\
&= 2\,(6-2) - (10-3) - (10-9) \\
&= 8 - 7 - 1 \\
&= 0
\end{aligned}
$$

The matrix is therefore not invertible.

Definition 4: The set of all $n \times n$, invertible matrices is called the general linear group of degree n.

Notation: $GL_n(\mathbb{R})$ denotes the general linear group of degree n with only real entries.

- $GL_n(\mathbb{R}) \coloneqq \{\, A \in M_n(\mathbb{R})\colon A \text{ is invertible} \,\}$.

Notation: $\mathbb{Z}_n$ denotes the set of all integers modulo n.

- $\mathbb{Z}_n \coloneqq \{\, 0, 1, \ldots, n-1 \,\}$.

Definition 5: $u \in \mathbb{Z}_n$ is called a unit modulo n if there exists some $v \in \mathbb{Z}_n$ such that $(uv) \mod n = 1$.

Notation: $\mathbb{Z}_n^\times$ is the set of all units modulo n.

Exercise 5: Write out the addition and multiplication tables for $\mathbb{Z}_{10}$. List all elements of $\mathbb{Z}_{10}^\times$.

Solution:

(I) Write out the addition table of $\mathbb{Z}_{10}$.

+	0	1	2	3	4	5	6	7	8	9
0	0	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9	0
2	2	3	4	5	6	7	8	9	0	1
3	3	4	5	6	7	8	9	0	1	2
4	4	5	6	7	8	9	0	1	2	3
5	5	6	7	8	9	0	1	2	3	4
6	6	7	8	9	0	1	2	3	4	5
7	7	8	9	0	1	2	3	4	5	6
8	8	9	0	1	2	3	4	5	6	7
9	9	0	1	2	3	4	5	6	7	8

(II) Write out the multiplication table of $\mathbb{Z}_{10}$.